Regenerative Braking System for Electric Vehicles: Architecture, Energy Recovery and Control Strategies

Author Name
Department of Electrical and Electronics Engineering
College/University Name, City, Country
Email: author@email.com

***Abstract*—Regenerative braking is an energy recovery technique used in electric vehicles (EVs) and hybrid electric vehicles (HEVs). It converts a portion of the kinetic energy available during deceleration into electrical energy using the traction motor operating as a generator. The recovered energy can be stored in the battery and reused for propulsion, potentially improving vehicle energy efficiency and driving range. This paper presents the architecture, operating principle, mathematical model, and control strategy of a regenerative braking system. The major components include the traction motor, bidirectional inverter, battery management system, electronic control unit, sensors, and conventional friction brakes. The paper also discusses braking torque coordination, energy conversion, advantages, limitations, and future developments. Safe operation requires coordination between regenerative and friction braking to maintain the required deceleration and vehicle stability under varying road and battery conditions.**

***Index Terms*—Regenerative braking, electric vehicle, energy recovery, traction motor, battery management, braking torque, power electronics.**

I. Introduction

Electric vehicles are increasingly used to reduce dependence on fossil fuels and improve transportation efficiency. However, a portion of the energy supplied to a vehicle is normally lost during braking. In conventional friction braking, the kinetic energy of the moving vehicle is converted mainly into heat at the brake discs and pads.

Regenerative braking provides an alternative by allowing the electric traction motor to operate as a generator during deceleration. Mechanical energy from the rotating wheels is converted into electrical energy and transferred through power electronics to the battery [1], [2].

The amount of recoverable energy depends on vehicle speed, mass, motor efficiency, battery state of charge, and road conditions. Regenerative braking cannot replace friction braking under all circumstances. A coordinated braking controller is therefore required to satisfy the driver's braking demand while recovering the maximum permissible energy.

This paper describes the system architecture, operating principle, mathematical relationships, control requirements, and limitations of regenerative braking in electric vehicles.

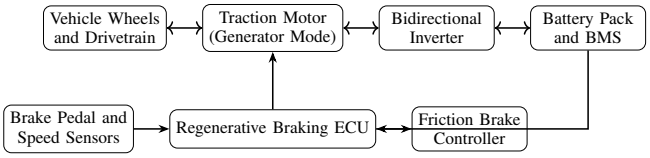


Fig. 1. Functional architecture of a regenerative braking system.

II. Related Work

Research on regenerative braking has investigated motor control, power converter design, braking torque distribution, battery charging limits, and vehicle stability. Conventional approaches use rule-based control to divide the required braking torque between regenerative and friction braking. More advanced strategies use optimization and predictive control to improve energy recovery while respecting safety constraints [1], [2].

The achievable energy recovery depends on both the electric drivetrain and the operating conditions. Battery charge acceptance may decrease at high state of charge or outside the preferred temperature range. At low vehicle speeds, regenerative power may also decrease, requiring friction brakes to complete the stop.

Consequently, regenerative braking should be designed as part of an integrated vehicle braking system rather than as an independent energy recovery device.

III. System Architecture

The main components of a regenerative braking system are the traction motor, bidirectional inverter, battery pack, battery management system (BMS), braking controller, sensors, and friction brakes. The motor is mechanically connected to the driven wheels. During braking, the wheels drive the motor as a generator and electrical energy flows toward the battery.

A. Functional Block Diagram

The sensors measure driver braking demand and vehicle operating conditions. The electronic control unit (ECU) estimates the available regenerative torque and sends commands to the motor controller. The BMS provides charging limits based on battery conditions. The friction brake controller supplies any remaining braking torque.

B. Major Components

Traction motor: Converts mechanical energy into electrical energy during generator operation.

Bidirectional inverter: Controls motor torque and manages electrical power transfer between the motor and DC bus.

Battery and BMS: Store recovered energy while enforcing voltage, current, temperature, and state-of-charge limits.

Braking ECU: Coordinates regenerative torque, friction braking, and vehicle safety inputs.

Sensors and friction brakes: Provide the operating measurements and supplementary braking required for safe vehicle deceleration.

IV. Methodology and Working Principle

A. Normal Driving

During acceleration, electrical energy flows from the battery through the inverter to the traction motor. The motor develops positive driving torque and transfers mechanical power to the wheels.

B. Regenerative Braking

When the driver requests deceleration, the controller determines whether regenerative braking is permitted. The motor develops torque opposing wheel rotation and operates as a generator. Generated electrical power passes through the inverter and is transferred to the battery when charging is allowed.

The operating sequence is:

1) Read brake-pedal and vehicle-speed signals.
2) Estimate requested braking torque.
3) Determine motor and battery regenerative limits.
4) Command generator torque through the inverter.
5) Apply friction braking for any torque deficit.
6) Monitor wheel slip and battery charging conditions.

C. Energy Flow Diagram

Not all kinetic energy is stored in the battery. Some energy is lost through aerodynamic drag, rolling resistance, electrical losses, and mechanical losses. Friction brakes may also dissipate part of the braking energy as heat.

V. Mathematical Modeling

A. Kinetic Energy

The translational kinetic energy of a vehicle is

$$E_k = \frac{1}{2}mv^2 \tag{1}$$

where m is vehicle mass in kilograms and v is speed in metres per second.

For a speed reduction from v_i to v_f, the decrease in kinetic energy is

$$\Delta E_k = \frac{1}{2}m(v_i^2 - v_f^2). \tag{2}$$

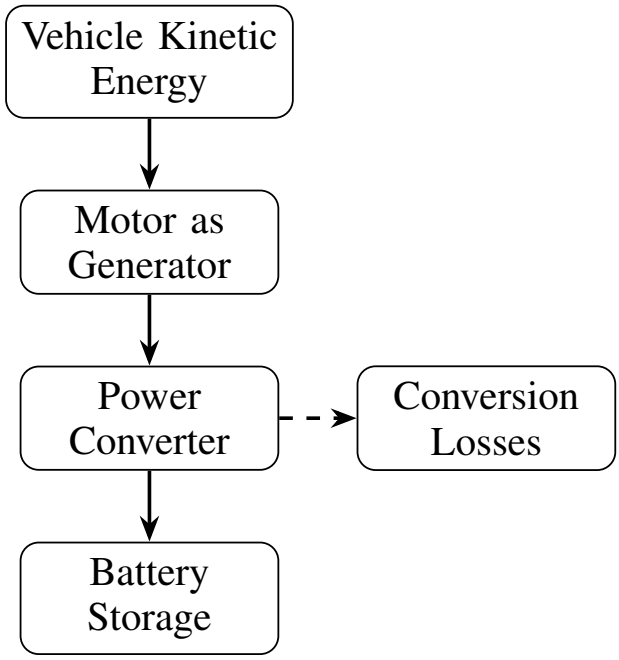


Fig. 2. Energy conversion during regenerative braking.

B. Recovered Energy

A simplified estimate of stored energy is

$$E_{\text{stored}} \approx \eta_{\text{path}} E_{\text{available}}, \tag{3}$$

where η_{path} represents the combined efficiency of the motor, converter, and battery charging path. The available energy is lower than the total kinetic energy reduction because of road loads and braking losses.

Average recovered power over a braking interval is

$$P_{\text{avg}} = \frac{E_{\text{stored}}}{t_b}, \tag{4}$$

where t_b is the braking duration.

C. Braking Torque Distribution

The total requested braking torque is divided between regenerative and friction braking:

$$T_{\text{req}} = T_{\text{reg}} + T_{\text{fric}}. \tag{5}$$

The controller restricts regenerative torque according to motor capability, battery charging power, and tire adhesion. If regenerative braking cannot satisfy the requested torque, friction braking supplies the difference. Actual systems must also account for axle torque distribution, drivetrain ratios, and wheel-slip constraints.

VI. Control Strategy

A. Torque Control

The ECU estimates braking demand using the brake-pedal signal and vehicle speed. It then calculates the maximum permitted regenerative torque. The inverter controls motor current to achieve the commanded torque.

B. Battery Management

The BMS communicates the battery's permitted charging power. When the battery is nearly full, too cold, too hot, or otherwise unable to accept sufficient charge, the controller reduces regeneration and increases friction braking as needed.

C. Vehicle Stability

On slippery roads, excessive regenerative torque can cause wheel slip at the driven axle. Coordination with the anti-lock braking system (ABS) and stability controller is therefore important. Regenerative torque must be reduced when necessary to preserve traction and directional control.

Rule-based strategies are comparatively simple, while optimization-based and predictive strategies can account for multiple operating constraints. All strategies require validation to ensure safe and predictable braking.

VII. Advantages and Limitations

A. Advantages

Regenerative braking can recover energy that would otherwise be dissipated as heat. The recovered energy may reduce net battery consumption and improve driving range, especially in driving cycles with frequent deceleration. Reduced use of friction brakes can also decrease brake wear and heat.

B. Limitations

Energy recovery is constrained by motor speed, conversion efficiency, battery charging limits, and tire-road adhesion. Regeneration is generally reduced at very low speeds and when the battery cannot accept additional charge. The system also requires coordinated control and reliable sensing. Regenerative braking does not eliminate the need for conventional friction brakes.

VIII. Applications and Future Scope

Regenerative braking is used in battery electric vehicles, hybrid vehicles, electric buses, and rail transportation. Future developments may combine predictive control, navigation data, improved battery charge-acceptance estimation, and adaptive torque distribution.

Model predictive control and other optimization methods may improve the balance between energy recovery, comfort, and stability. Any improvement should be evaluated through simulation and controlled testing using measurements such as recovered energy, stopping distance, braking response, and wheel-slip behaviour.

IX. Conclusion

Regenerative braking is an important energy recovery technology for electric and hybrid vehicles. It enables the traction motor to convert a portion of the vehicle's kinetic energy into electrical energy during deceleration. The architecture combines the motor, inverter, battery, sensors, braking ECU, and friction brakes.

The performance of the system depends on motor capability, battery charging limits, vehicle speed, and road conditions. A coordinated braking strategy is essential to meet the driver's braking demand while preserving vehicle stability. Further improvements in predictive control and battery management may increase energy recovery, but safety and reliable braking performance must remain the primary design requirements.

Acknowledgment

The author acknowledges the technical literature and engineering principles used to develop this conceptual review of regenerative braking systems.

References

[1] E. M. Szumska, "Regenerative braking systems in electric vehicles: A comprehensive review of design, control strategies, and efficiency challenges," *Energies*, vol. 18, no. 10, Art. no. 2422, May 2025, doi: 10.3390/en18102422.

[2] P. Saiteja, B. Ashok, A. S. Wagh, and M. E. Farrag, "Critical review on optimal regenerative braking control system architecture, calibration parameters and development challenges for EVs," *International Journal of Energy Research*, vol. 46, no. 14, pp. 20146–20179, 2022, doi: 10.1002/er.8306.

[3] J. Zhang *et al.*, "Fully electrified regenerative braking control for deep energy recovery and maintaining safety of electric vehicles," *IEEE Transactions on Vehicular Technology*, vol. 65, no. 3, pp. 1186–1198, Mar. 2016, doi: 10.1109/TVT.2015.2410694.

[4] "Modeling and integration of electric vehicle regenerative and friction braking for motor/dynamometer test bench emulation," *IEEE Transactions on Vehicular Technology*, vol. 65, no. 6, pp. 4264–4273, Jun. 2016, doi: 10.1109/TVT.2015.2504363.